

A systematic screening of total antioxidants in dietary plants.

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A predominantly plant-based diet reduces the risk for development of several chronic diseases. It is often assumed that antioxidants contribute to this protection, but results from intervention trials with single antioxidants administered as supplements quite consistently do not support any benefit. Because dietary plants contain several hundred different antioxidants, it would be useful to know the total concentration of electron-donating antioxidants (i.e., reductants) in individual items. Such data might be useful in the identification of the most beneficial dietary plants. We have assessed systematically total antioxidants in a variety of dietary plants used worldwide, including various fruits, berries, vegetables, cereals, nuts and pulses. When possible, we analyzed three or more samples of dietary plants from three different geographic regions in the world. Total antioxidants was assessed by the reduction of $\text{Fe}(3+)$ to $\text{Fe}(2+)$ (i.e., the FRAP assay), which occurred rapidly with all reductants with half-reaction reduction potentials above that of $\text{Fe}(3+)/\text{Fe}(2+)$. The values, therefore, expressed the corresponding concentration of electron-donating antioxidants. Our results demonstrated that there is more than a 1000-fold difference among total antioxidants in various dietary plants. Plants that contain most antioxidants included members of several families, such as Rosaceae (dog rose, sour cherry, blackberry, strawberry, raspberry), Empetraceae (crowberry), Ericaceae (blueberry), Grossulariaceae (black currant), Juglandaceae (walnut), Asteraceae (sunflower seed), Punicaceae (pomegranate) and Zingiberaceae (ginger). In a Norwegian diet, fruits, berries and cereals contributed 43.6%, 27.1% and 11.7%, respectively, of the total intake of plant antioxidants. Vegetables contributed only 8.9%. The systematic analysis presented here will facilitate research into the nutritional role of the combined effect of antioxidants in dietary plants.